

CYBERSECURITY AWARENESS WORKSHEET

Spot the Secure Website: HTTP vs HTTPS

Worksheet completed by Sam

This activity helps participants identify secure vs insecure websites by analysing URL structure, browser indicators, and trust signals.

The goal is to improve awareness of how to recognise potentially unsafe websites and understand that HTTPS alone does not guarantee legitimacy.

Section 1: Website Security Evaluation

Website	URL	Indicators	Secure? (Yes/No)
A	http://online-banking-login.example.com	No padlock, HTTP	No
B	https://university.edu/login	Padlock, HTTPS	Yes
C	https://secure-paypal-login.example.net	HTTPS but suspicious domain	No

Questions:

Which website is most secure? Why?

Website B, as it uses HTTPS encrypted communication between a client and a web server.

Why is HTTP considered unsafe?

HTTP doesn't use encrypted communication between endpoints, and is therefore susceptible to interception attacks.

Does HTTPS always mean a site is trustworthy? Explain.

HTTPS only provides encrypted communication between endpoints, and doesn't guarantee that the website is actually safe.

Section 2: Safe vs Unsafe Links

Evaluate the following links-

Task:

- Mark each link as Safe / Suspicious
- Identify one red flag for each suspicious link

1. <https://paypal.com/login>

Safe, as it is a standard website endpoint.

2. <https://paypal-secure-login.verify-now.example.com>

Suspicious, as it has different subparts within the URL.

3. <https://uwa.edu.au>

Safe, as it appears as a trusted website.

4. <http://free-prizes.example.net>

Suspicious, as includes an unreliable and unrealistic free prizes section.

Section 3: File Safety (Malware Awareness)

Determine whether these files are safe:

- invoice.pdf.exe
- report.docx
- update_patch_v2.zip
- photo.png

Questions:

- Which file is most suspicious? Why?
- What is a common trick used in malicious file names?

Invoice.pdf.exe, as it includes a double file extension.

This may trick the user into clicking on a seemingly safe word document, and actually having malicious code execute on their computer.

Double file extension, or unsafe naming conventions is a common trick.

Section 4: Password Strength Analysis

Rank the following passwords from weakest → strongest:

- ☐1 123456
- ☐3 password2024
- ☐2 sam12345
- ☐4 UWA!secure9
- ☐5 Xy9#kL2pQ

Questions:

- What makes a password strong?
- Why is password reuse dangerous?

Strong passwords have length, randomness, and special characters.

Password reuse compromises other accounts, if attackers know the common password.

Section 5: Social Engineering Scenarios

Scenario 1:

You receive a message from “IT Support” asking for your password to fix an issue.

Scenario 2:

A friend messages you urgently asking for money.

Questions:

- What are the warning signs?
- What would you do in each situation?

Scenario 1 – Warning sign is asking for your private password.

I would contact the actual IT support service to verify this.

Scenario 2 – Warning sign is urgency and emotional attack.

I would contact this person on another platform of communication to verify this.

Section 6: Quick Knowledge Check (Mixed Questions)

Multiple Choice:

1. What does HTTPS provide?
 - ☒ Encryption
 - ☐ Faster internet speed
 - ☐ Virus protection
2. What is phishing?
 - ☐ A type of malware
 - ☒ A social engineering attack
 - ☐ A firewall

True / False:

- HTTPS guarantees a website is completely safe (T / F) F
- Strong passwords should include symbols and numbers (T / F) T

Section 4: Reflection

What new cybersecurity concept did you learn?

Learnt about insecurity with files containing double extensions.

Which section did you find most difficult?

The URL/website analysis part, as it can be ambiguous.

How will this activity change your online behaviour?

I will verify unreliable claims with primary sources upon the detection of dodgy messages.
